Cell Tuition
1-1 PRIVATE TUTORS

NUMBER

Divisibility Rules

Name _____ Date _____ No calculator Version 1

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KEY WORDS divisible · digit sum · remainder *(toenga)* · multiple *(tau whakarea)*

A Dividing by 2, 5 and 10 ●○○

Remember: Look at the last digit. Divisible by 2: it is even (0, 2, 4, 6, 8). Divisible by 5: it is 0 or 5. Divisible by 10: it is 0.

1. Circle the numbers in each list.

a) Divisible by 2: 14 27 50 63 98 105 236 1,001

b) Divisible by 5: 15 32 40 54 75 101 230 555

c) Divisible by 10: 20 35 70 102 150 205 900 1,010

B Dividing by 3, 6 and 9 ●●○

Remember: Add the digits. Divisible by 3: the digit sum is divisible by 3. Divisible by 9: the digit sum is divisible by 9. Divisible by 6: the number is even and divisible by 3.

2. Is each number divisible by 3? Show the digit sum.

a) 42 **b)** 71 **c)** 135

d) 208 **e)** 561 **f)** 1,004

3. Is each number divisible by 9? Show the digit sum.

a) 54 **b)** 93 **c)** 171

d) 288 **e)** 406 **f)** 1,233

4. Is each number divisible by 6? Give a reason.

a) 48 **b)** 81 **c)** 124

d) 210 **e)** 333 **f)** 504

C Dividing by 4 and 8 ●●●

Remember: Divisible by 4: the number made by the last two digits is divisible by 4. Divisible by 8: the number made by the last three digits is divisible by 8.

5. Is each number divisible by 4? Give a reason.

a) 116 **b)** 230 **c)** 524

d) 1,318 **e)** 2,736 **f)** 5,008

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6. Is each number divisible by 8? Give a reason.

a) 1,120 **b)** 3,204 **c)** 5,016

d) 7,100

D All the rules together ●●●

7. Put a tick in each box where the number divides exactly.

Number	÷ 2	÷ 3	÷ 4	÷ 5	÷ 6	÷ 9	÷ 10
120							
315							
432							
1,005							
728							
2,340							

E Problems ●●●

8. Use a divisibility rule to answer each question. Give a reason.

a) 132 students are put into teams of 6. Will every student be in a full team?

b) The same 132 students are put into groups of 4. Will every student be in a full group?

c) 250 eggs are packed into boxes of 6. Will all the boxes be full?

d) 207 lollies are shared equally between 9 children. Will there be any left over?

★ Extension

9. Which digits could go in the box? Find them all.

a) 4☐2 is divisible by 3 **b)** 57☐ is divisible by 4

c) 3☐6 is divisible by 9 **d)** 82☐ is divisible by 6

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1-1 PRIVATE TUTORS

NUMBER

Divisibility Rules

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS divisible · digit sum · remainder *(toenga)* · multiple *(tau whakarea)*

A Dividing by 2, 5 and 10 ●○○

Remember: Look at the last digit. Divisible by 2: it is even (0, 2, 4, 6, 8). Divisible by 5: it is 0 or 5. Divisible by 10: it is 0.

1. Circle the numbers in each list.

a) Divisible by 2: 16 29 30 47 84 107 352 2,001

b) Divisible by 5: 25 38 60 52 85 103 410 505

c) Divisible by 10: 30 45 80 104 250 305 700 2,020

B Dividing by 3, 6 and 9 ●●○

Remember: Add the digits. Divisible by 3: the digit sum is divisible by 3. Divisible by 9: the digit sum is divisible by 9. Divisible by 6: the number is even and divisible by 3.

2. Is each number divisible by 3? Show the digit sum.

a) 51 **b)** 82 **c)** 147

d) 305 **e)** 654 **f)** 1,003

3. Is each number divisible by 9? Show the digit sum.

a) 63 **b)** 84 **c)** 189

d) 279 **e)** 505 **f)** 2,133

4. Is each number divisible by 6? Give a reason.

a) 54 **b)** 93 **c)** 136

d) 312 **e)** 555 **f)** 702

C Dividing by 4 and 8 ●●●

Remember: Divisible by 4: the number made by the last two digits is divisible by 4. Divisible by 8: the number made by the last three digits is divisible by 8.

5. Is each number divisible by 4? Give a reason.

a) 128 **b)** 350 **c)** 716

d) 1,422 **e)** 3,548 **f)** 6,004

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6. Is each number divisible by 8? Give a reason.

a) 1,160 **b)** 2,308 **c)** 4,024

d) 9,100

D All the rules together ●●●

7. Put a tick in each box where the number divides exactly.

Number	÷ 2	÷ 3	÷ 4	÷ 5	÷ 6	÷ 9	÷ 10
180							
225							
624							
1,010							
936							
4,320							

E Problems ●●●

8. Use a divisibility rule to answer each question. Give a reason.

a) 156 students are put into teams of 6. Will every student be in a full team?

b) The same 156 students are put into groups of 4. Will every student be in a full group?

c) 340 eggs are packed into boxes of 6. Will all the boxes be full?

d) 315 lollies are shared equally between 9 children. Will there be any left over?

★ Extension

9. Which digits could go in the box? Find them all.

a) 5☐1 is divisible by 3 **b)** 63☐ is divisible by 4

c) 2☐7 is divisible by 9 **d)** 74☐ is divisible by 6

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